

ONBOARDING FLOW — RUNS ONCE AT SIGNUP

- 1

Sign in with Google OAuth2

Passport.js consent — scopes: Gmail read + compose + send. Tokens AES-256-GCM encrypted before DB write.
- 2

Pull last 30 sent emails

Gmail API label:SENT. Fetched once, never stored. Passed directly to Claude for analysis.
- 3

Claude analyses writing style

Extracts formality score, avg sentence length, common openers/closers, punctuation and emoji habits.
- 4

Show inferred profile to user

"You write semi-formally in short sentences." User confirms or adjusts default tone and signature.
- 5

Save profile — onboarding complete

preferences JSONB updated. onboardingComplete = true, onboardingAt = now(). Redirect to inbox.

EMAIL DRAFTING FLOW — PER INCOMING EMAIL

- 1

New email arrives

Polling or Gmail push webhook
- 2

Hard filter (free — no AI)

Skip: no-reply · PROMOTIONS label · List-Unsubscribe header
- 3

Claude triage — in memory, then discarded

Classify: actionable · informational · notification · automated
- 4

Load style profile from DB

users.preferences — no re-fetch of sent mail needed
- 5

Claude generates draft

Style profile + thread context + signature + promptVersion
- 6

Save to email_drafts (status = pending)

priority from triage · idempotency key = hash(userId+messageId)
- 7

User reviews — tone override available

Friendly · Formal · Concise on top of baseline profile
- 8

Approve / Edit+Approve / Reject

status updated · edit diff stored · rejected drafts never sent
- 9

Send via Gmail API + write audit log

In-Reply-To + References headers · BullMQ retry on failure

DATABASE SCHEMA — 3 TABLES ONLY

users	
id	UUID PK
googleId	VARCHAR unique
email	VARCHAR
encryptedTokens	TEXT (AES-256-GCM)
preferences	JSONB
onboardingComplete ★	BOOLEAN
onboardingAt ★	TIMESTAMP TZ
profileReviewedAt ★	TIMESTAMP TZ

email_drafts	
id	UUID PK
userId	UUID FK → users
messageId	VARCHAR
threadId	VARCHAR
status	ENUM pending approved edited rejected sent
priority ★	ENUM high medium low
aiDraft	TEXT
finalDraft	TEXT
promptVersion	VARCHAR
createdAt	TIMESTAMP TZ

audit_logs	
id	UUID PK
userId	UUID FK → users
draftId	UUID FK → email_drafts
action	VARCHAR
gmailMessageId	VARCHAR
createdAt	TIMESTAMP TZ

★ New fields — no separate triage_results table needed.

30-DAY TONE PROFILE EVOLUTION

- Style profile stays current without surprising the user

1

BullMQ cron fires every 30 days

Checks NOW() - profileReviewedAt > 30 days per user. Runs silently in background.

2

Pull last 30 sent emails again

Same Gmail API call as onboarding — fresh snapshot of current writing habits.

3

Claude computes delta vs stored profile

If drift below threshold — skip, no update. Only significant changes proceed.

4

Nudge user — never update silently

In-app banner: "Your style seems to have shifted — update your profile?" User accepts or dismisses.

5

On accept — update preferences + profileReviewedAt

PATCH users.preferences with new styleProfile JSON. All future drafts use refreshed profile.

TECH STACK

BACKEND Node.js 20 + Express + TS	AUTH Passport.js OAuth2 + Redis
DATABASE PostgreSQL + Prisma ORM	QUEUE / CACHE Redis + BullMQ (cron)
AI MODEL Anthropic Claude API	FRONTEND React 18 + Vite + TanStack
SECURITY Helmet · Zod · AES-256-GCM	DEPLOY Docker + Railway